

# CURRICULUM VITAE

## PERSONAL INFORMATION

**Name:** Anjuman Ara Khatun, PhD  
**Degrees:** Doctor of Philosophy (PhD)  
**Current address:** Kofuchu-cho, Kofu, Yamanashi, Japan  
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## PROFESSIONAL OBJECTIVE

To obtain a research and academic position where I can effectively apply my educational background, research expertise, and innovative thinking to advance scientific knowledge, contribute to impactful research, and support excellence in teaching and mentoring. I aspire to contribute to the growth of the institution while addressing global challenges through high-quality research and academic activities.

### Research expertise & core competencies

- Biomedical Engineering, Polymer Materials, and Biodegradable Polymers
- Density Functional Theory (DFT) calculations and computational materials research
- Synthesis, processing, and characterization of ceramic materials, metal alloys, and composite materials
- Advanced joining technologies, including Electron Beam Welding and Laser Welding
- Scientific writing, peer-reviewed publication, and research dissemination
- Student supervision, academic mentoring, and teaching support
- Research proposal development and grant writing
- Experimental design, data analysis, and engineering problem-solving
- Proficient in operating characterization instruments and scientific software for materials analysis
- Manuscript preparation, technical reporting, and research project management
- Strong leadership, teamwork, critical thinking, communication, and time-management skills
- Dedicated researcher with a high level of professionalism, integrity, responsibility, and commitment to academic excellence

## ACADEMIC QUALIFICATIONS

- |                                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|---------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Mar 2024</b><br>Yamanashi,<br>Japan      | <b>Doctor of Philosophy</b> , in Energy Materials Science, Department of Engineering<br>University of Yamanashi, Japan <ul style="list-style-type: none"><li>• <b>Supported:</b> <i>Fully Funded by Japanese Government MEXT Scholarship, Japan</i></li><li>• <b>Thesis title:</b> Study on solid solution and characterization of substituted niobate columbite with di- and trivalent cations</li><li>• <b>Result:</b> 3.60 out of 4.00</li></ul>                                              |
| <b>Sep 2018</b><br>Rajshahi,<br>Bangladesh  | <b>Master of Science</b> , Materials science and Engineering, University of Rajshahi, Bangladesh <ul style="list-style-type: none"><li>• <b>Supported:</b> <i>Fully Funded Fellowship by Bangladesh's Ministry of Science and Technology.</i></li><li>• <b>Thesis title:</b> Synthesis and characterization of pH responsive poly (Maleic acid-co-Malic acid-co-Butane-1,4-diol-co-Adipic acid)</li><li>• <b>Result:</b> 3.65 out of 4.00 (<i>Certificate of Excellence from Dean</i>)</li></ul> |
| <b>Aug 2016</b><br>Rajshahi,<br>Bangladesh  | <b>Bachelor of Science</b> , Materials science and Engineering, University of Rajshahi, Bangladesh <ul style="list-style-type: none"><li>• <b>Supported:</b> Funded by <i>University Scholarship</i></li><li>• <b>Research Project title:</b> Nanotechnology for pollution control</li><li>• <i>Certificate of Excellence from Dean</i></li></ul>                                                                                                                                                |
| <b>July 2009</b><br>Dinajpur,<br>Bangladesh | <b>Higher Secondary Certificate</b> , Dinajpur Government College, Dinajpur. <ul style="list-style-type: none"><li>• <b>Supported:</b> <i>Government Merit Scholarship</i></li><li>• <b>Result:</b> 5.00 out of 5.00 (<i>Awarded for outstanding academic performance</i>)</li></ul>                                                                                                                                                                                                             |
| <b>June 2007</b><br>Dinajpur,<br>Bangladesh | <b>Secondary School Certificate</b> , Dinajpur Government Girls High School, Dinajpur. <ul style="list-style-type: none"><li>• <b>Supported:</b> <i>Government Merit Scholarship</i></li><li>• <b>Result:</b> 5.00 out of 5.00 (<i>Awarded for outstanding academic performance</i>)</li></ul>                                                                                                                                                                                                   |

## RESEARCH AND WORK EXPERIENCE

|                                                    |                                                                                                                                                                       |
|----------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Nov 2024 – Present</b><br>Yamanashi, Japan      | <b>Research and Development Engineer</b> , Nakaya Co. Ltd, Yamanashi, Japan                                                                                           |
| <b>Apr 2024 - Sep 2024</b><br>Yamanashi, Japan     | <b>Postdoctoral Researcher</b> , Center for Crystal Science and Technology, University of Yamanashi, Japan                                                            |
| <b>Sep 2024 – Nov 2024</b><br>Yamanashi, Japan     | <b>Lecturer in Mathematics</b> , International School-The American School in Kofu, Japan                                                                              |
| <b>Oct 2021 – Mar 2024</b><br>Yamanashi, Japan     | <b>Research Student &amp; Advisor for Junior Student</b> , Department of Energy Materials Science, University of Yamanashi, Japan                                     |
| <b>Apr 2021 – Sep 2023</b><br>Yamanashi, Japan     | <b>Student Assistant (Lecturer)</b> , Center for Language Education (English Café), University of Yamanashi, Japan                                                    |
| <b>Oct 2018-Mar 2021</b><br>Dhaka, Bangladesh      | <b>Research Student &amp; Advisor for Junior Students</b> , Nanomaterials and Ceramic Engineering, BUET, Bangladesh                                                   |
| <b>Oct 2018-Mar 2021</b><br>Dhaka, Bangladesh      | <b>Sub-editor</b> , Daily English newspaper in ‘The Asian Age, Dhaka, Bangladesh                                                                                      |
| <b>Janu 2018-Sep 2018</b><br>Rajshahi, Bangladesh  | <b>Research Assistant &amp; Advisor for Undergraduate Students</b> , Simulation and Functional Ceramic Lab, Materials Science and Engineering, University of Rajshahi |
| <b>Mar 2018-Sep 2018</b><br>Dhaka, Bangladesh      | <b>Branch Manager</b><br>Sunshine Insurance Company Ltd., Bangladesh                                                                                                  |
| <b>Sep 2016 – Sep 2018</b><br>Rajshahi, Bangladesh | <b>Research Student &amp; Advisor for Undergraduate Students</b> , Eco-Friendly and Biodegradable Polymeric Materials Lab, University of Rajshahi, Bangladesh         |
| <b>Aug 2015-Sep 2018</b><br>Rajshahi, Bangladesh   | <b>Lecturer</b> , Romen English Academy, Rajshahi, Bangladesh                                                                                                         |
| <b>Oct 2018 -Feb 2021</b><br>Rajshahi, Bangladesh  | <b>Lecturer</b> , Teaching experience as an English, Chemistry and ICT teacher in Coaching Center, Dhaka, Bangladesh                                                  |

### Volunteer/substitute teacher

- ✓ Yamanashi Prefectural Kofu First High School, Japan (2,10 February, 2024)
- ✓ Retina Coaching Centre, Bangladesh (10 December, 2017)
- ✓ Rajshahi University Higher Study Club (13-15 June 2014)
- ✓ Volunteer in Blood donation club of Rajshahi University.

## PEER-REVIEWED SCIENTIFIC PUBLICATIONS & EXPECTED TO PUBLICATIONS

- [15]. Muhamad Diki Permana, Takahiro Takei, **Anjuman Ara Khatun**, Norio Saito and Nobuhiro Kumada, The synergistic effect of Fe<sub>3</sub>O<sub>4</sub>/g-C<sub>3</sub>N<sub>4</sub> for persulfate activation in sulfate radical photo-Fenton, Colloids and Surfaces A Physicochemical and Engineering Aspects Volume 720, 5 September 2025, 137110 <https://doi.org/10.1016/j.colsurfa.2025.137110>
- [14]. Shojib Mia, Firoz Ahmed, Ibrahim Khan, Md Imamul Kabir, Mehedi H. Roni, Khadijatul Cobra, **Anjuman Ara Khatun**, Shahriar Mahmud, Integrating Renewable Energy with IoT: Pathways to a Smart Green Planet, *Kufa Journal of Engineering*, 16(1), 361-404 (2025), <https://doi.org/10.30572/2018/KJE/160121>
- [13]. Abu Mahmud, **Anjuman Ara Khatun**, M. Abdul Matin, Monishita Nuzhat Chowdhury, M. Asadul Hoque, pH-Sensitive Biodegradability of Poly(Maleic acid-co-Malic acid-coButane-1,4-diol-co-Adipic acid) for Smart Release of Fertilizers, *Journal of Composites and Biodegradable Polymers*, Vol. 1234–40 (2024) <https://doi.org/10.12974/2311-8717.2024.12.04>
- [12]. Muhamad Diki Permana, Takahiro Takei, **Anjuman Ara Khatun**, Norio Saito and Nobuhiro Kumada, Effect of wavelength in light irradiation for Fe<sup>2+</sup>/Fe<sup>3+</sup> redox cycle of Fe<sub>3</sub>O<sub>4</sub>/g-C<sub>3</sub>N<sub>4</sub> in photocatalysis and photo Fenton systems, *Journal of photochemistry and photobiology A: chemistry*, 457, 115876 (2024) <https://doi.org/10.1016/j.jphotochem.2024.115876>

- [11]. Rukaia Khatun, Md. Atikur Rahman Atik, Dayal Chandra Roy, **Anjuman Ara Khatun**, Mukter Hossain, Upasana Rani, Peeyush Kumar Kamlesh, Ahmad Irfan, Sarah Chaba Mouna, DFT study on the structural, mechanical, electronic, optical and thermodynamic properties of recently synthesized MAX Phase compounds  $A_3\text{InC}_2$  ( $A = \text{Zr, Hf}$ ) under ambient and elevated pressure, Materials today communication Volume 40, August 2024, 109964, <https://doi.org/10.1016/j.mtcomm.2024.109964>
- [10]. **Anjuman Ara Khatun**, Takahiro Takei, Muhamad Diki Permana, Norio Saito, Nobuhiro Kumada, Effect of silver hybridization on magnesium and calcium niobate: properties, photocatalytic and antibacterial activity, Journal of society of Inorganic Materials, Japan, 31, 119-126 (2024), [https://doi.org/10.11451/mukimate.31.430\\_119](https://doi.org/10.11451/mukimate.31.430_119)
- [9]. **Anjuman Ara Khatun**, Takahiro Takei, Muhamad Diki Permana, Norio Saito, Nobuhiro Kumada, Study on Preparation and Characterization of Orthorhombic Euxenite-Type Solid Solutions between  $\text{TbNbTiO}_6$  and  $\text{DyNbTiO}_6$ , Journal of the Ceramic Society of Japan, 132 (4), 178-183 (2024) <https://doi.org/10.2109/jcersj2.23182>
- [8]. **Anjuman Ara Khatun**, Takahiro Takei, Muhamad Diki Permana, Norio Saito, Nobuhiro Kumada, Synthesis and characterization of columbite type solid solution structured niobate  $\text{Mg}_{1-x}\text{Ca}_x\text{Nb}_2\text{O}_6$ , Journal of the Ceramic Society of Japan, 131(8), 355-362 (2023) <https://doi.org/10.2109/jcersj2.23026>
- [7]. M. Abdul Matin, Monishita Nuzhat Chowdhury, **Anjuman Ara Khatun**, Abu Mahmud, Jahanara Nasrin, Abu Mahmud, Synthesis and Characterization of Stimuli-Responsive Poly (Maleic Acid-Co-Malic Acid-Co-Propane-1,2-Diol-Co-Adipic Acid), European Journal of Applied Sciences, 11(2), 2023, <https://doi.org/10.14738/aivp.112.14193>
- [6]. Khandaker Monower Hossain, S. K. Mitro, Mohammad Mizanur Rahman, **Anjuman Ara Khatun**, F. Parvin, Analyzing the physical properties of perovskite oxides  $\text{BaMO}_3$  ( $M = \text{Ru, Os}$ ) for predicting potential applications, Computational Condensed Matter, 34, e00782 (2023), <https://doi.org/10.1016/j.cocom.2022.e00782>
- [5]. M. H. K. Rubel, M. A. Hossain, M. Khalid Hossain, K. M. Hossain, **A. A. Khatun**, M. M. Rahaman, Md. Ferdous Rahman, M.M. Hossain, First-principles calculations to investigate structural, elastic, electronic, thermodynamic, and thermoelectric properties of  $\text{CaPd}_3\text{B}_4\text{O}_{12}$  ( $B = \text{Ti, V}$ ) perovskites, Results in Physics, Volume 42, November 2022, 105977 <https://doi.org/10.1016/j.rinp.2022.105977>
- [4]. Md. Mijanur Rahaman, Mirza H. K. Rubel, Md. Abdur Rashid, M. Ashraful Alam, Khandaker Monower Hossain, Md. Imran Hossain, **Anjuman Ara Khatun**, Md. Mukter Hossain, A. K. M. Azharul Islam, Seiji Kojima and Nobuhiro Kumada, Mechanical, electronic, optical and thermodynamic properties of orthorhombic  $\text{LiCuBiO}_4$  crystal: A first-principles study, Journal of Materials Research and Technology, 8(5), 3783-3794 (2019), <https://doi.org/10.1016/j.jmrt.2019.06.039>
- [3]. M. N. Hossain, M. D. Islam, A. Rahaman, M. A. Hena, **A. A. Khatun** and M. A. Matin, Removal of chromium from tannery effluents using egg and snail shells, Proc. Of 1st International Conference on Sustainability in Natural and Built Environment (iCSNBE 2019), Dhaka, Bangladesh, ISBN: 978-0-6482681-4-7 (2019), [Removal of chromium from tannery effluents using egg and snail shells](#)
- [2]. M.R. Hossain, **A.A. Khatun**, M.F. Ahmed, M.O. Faroque & M.A. Sobahan, Experimental Behavior of Bituminous Mixes with Waste Concrete Aggregate, International Journal of Educational Researchers, 9(3), 32-50 (2018).
- [1]. M. Abdul Matin, Abu Mahmud, M. Nurul Islam, **Anjuman Ara Khatun** and Monishita Nuzhat Chowdhury, Polymerization Kinetics, Biodegradation and pH-Responsive Behavior of Poly [(Citric Acid-co-PEG-400)-Ran-(Glycerol)], Journal of Composites and Biodegradable Polymers, 2018, 6, 20-27, [10.12974/2311-8717.2018.06.3](https://doi.org/10.12974/2311-8717.2018.06.3)

#### ON GOING PROJECT

1. **Anjuman Ara Khatun**, Khan, Muhammad Abid, Solomon wakesa, A comprehensive review of electron beam welding of aluminum alloys 5052 and 6061 metallurgical behavior, mechanical properties, and industrial applications (**Under review**).
2. Khan Muhammad Abid, **Anjuman Ara Khatun**, Koji Dairaku, A Validated Reanalysis-Based and Geospatial Framework for Climate-Relevant Wind Energy Assessment and Planning in Pakistan, Journal information; Quartiles no.1, Impact factor 09 (**Under review**)
3. **Anjuman Ara Khatun**, Takahiro Takei, Muhamad Diki Prmana, Norio Saito and nobuhiro Kumada, Effect of Titanium and Tungsten on Magnesium and Calcium Niobate: Properties, Photocatalytic activity (**In preparation**)

4. **Anjuman Ara Khatun** Optimizing EBW Parameter Settings and Welding Modes for Efficient and Perfect Welding (**In preparation**).
5. **Anjuman Ara Khatun** Sustainable Development of Biodegradable Disposable Tableware from Banana Biomass: A Review (**In preparation**)
6. Transforming Livelihoods Through Training, Research, Innovation, and Entrepreneurship for Sustainable Development (**Supervising work, currently going on**)

#### CONFERENCES, SEMINARS, WORKSHOP AND TRAINING

- |      |                                                                                                                                                                        |
|------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2023 | 61st Symposium on Basic Science of Ceramics, Okayama University, Japan, 2023 (Oral presentation)                                                                       |
| 2023 | International Conference on Electronics and Informatics, Atomic Energy Commission, Bangladesh, 2023 (Oral presentation)                                                |
| 2024 | University-Company Joint Workshop at the University of Yamanashi, Japan                                                                                                |
| 2023 | 61st Symposium on Basic Science of Ceramics, Okayama University, Japan, 2023 (Oral presentation)                                                                       |
| 2023 | International Conference on Electronics and Informatics, Atomic Energy Commission, Bangladesh, 2023 (Oral presentation)                                                |
| 2019 | International Conference on Sustainability in Built and Natural Environment, Dhaka, Bangladesh, 2019 (Attendee)                                                        |
| 2018 | International Conference on Advances in Computing, Electronics and Communication, Kuala Lumpur, Malaysia, 2018 (Oral presentation)                                     |
| 2018 | Inter University Scientific Poster Presentation Competition BUET, Bangladesh, 2018 (Poster presentation)                                                               |
| 2017 | International conference on computer, communication, chemical, materials, electronics and information engineering, University of Rajshahi, Bangladesh, 2017 (Attendee) |
| 2017 | Thermal analysis of materials by Differential Scanning Calorimetry (DSC), University of Rajshahi, Bangladesh, 2017 (Attendee)                                          |
| 2016 | Conference on Materials science and Nano-electrochemistry. (Attendee)                                                                                                  |
| 2016 | Seminar on materials for everyday life, Bangladesh, 2016.                                                                                                              |
| 2016 | Workshop on Self-Assessment Program and Art of Presentation, University of Rajshahi, Bangladesh.                                                                       |
| 2015 | Seminar on higher study in the world, Din's complex, University of Rajshahi, Bangladesh.                                                                               |
| 2014 | Completed a training from Fire service of Rajshahi about to save the people different types of danger.                                                                 |

#### CONTRIBUTION TO RESEARCH COMMUNITY

##### Editorial Board Membership:

- **Editorial Board Member:** *International Journal of MATERIAL SCIENCE*

##### Peer-Review Activities

- **Reviewer** for *Journal of Materials science & engineering. A* 2026
- **Reviewer** for *Journal of Materials science & engineering. A* 2026
- **Reviewer** for *Journal of Materials science & engineering. A* 2026
- **Reviewer** for *American Journal of Materials Synthesis and Processing* 2025
- **Reviewer** for *Journal of Engineering Research and Reviews* 2025
- **Reviewer** for *Journal of Engineering Research and Reviews* 2025

##### Supervisory Roles: Co-Supervisor, B.Sc. Graduation Projects:

- Artificial Intelligence for Predictive Maintenance in Manufacturing Industry, Jomo Kenyatta University of Agriculture and Technology, Kenya
- DFT- Based study of  $Ba_2MZO_6$  ( $M=La, Y+Z=Bi,Sb$ ), Pabna University of Science and Technology, Bangladesh.
- DFT calculation Perovskites, University of Rajshahi, Bangladesh.

## ACADEMIC SCHOLARSHIPS, FELLOWSHIPS & AWARDS

|      |                                                                                |
|------|--------------------------------------------------------------------------------|
| 2020 | Japanese Government Scholarship (MEXT) for Ph.D. study.                        |
| 2015 | Bangladesh Government Fellowship (NST) for Masters (M.Sc) study.               |
| 2011 | University Scholarship for B.Sc study from University of Rajshahi (Bangladesh) |
| 2015 | University Scholarship for M.Sc study from University of Rajshahi (Bangladesh) |
| 2009 | Dinajpur Board Scholarship for the result in H.S.C Examination. (Bangladesh)   |
| 2007 | Rajshahi Board Scholarship for the result in S.S.C. Examination. (Bangladesh)  |
| 2005 | Government Scholarship in class Eight (Bangladesh).                            |
| 2003 | Mathe Olympiad win in the district from School                                 |
| 2004 | Awards from extracurricular activities in school                               |
| 2012 | Different types of awards from extra-curricular activities in university       |

## KNOWLEDGE & SKILLS

- Extensive experience in the synthesis of inorganic noble compounds using various preparation methods.
- Strong expertise in the synthesis of nanomaterials through multiple chemical and physical techniques.
- Proven experience in the fabrication of composite materials using different synthesis approaches.
- Proficient in advanced characterization techniques, including X-ray Diffraction (XRD), Scanning Electron Microscopy (SEM) with Energy Dispersive Spectroscopy (EDS), Raman Spectroscopy, Fourier Transform Infrared Spectroscopy (FTIR), Thermogravimetric and Differential Thermal Analysis (TG-DTA), Transmission Electron Microscopy (TEM), X-ray Absorption Fine Structure (XAFS), and UV–Visible Spectroscopy etc.
- Experienced in Density Functional Theory (DFT) calculations for material property analysis.
- Hands-on experience in the synthesis and characterization of polymeric compounds.
- Skilled in using various scientific software tools for data analysis and interpretation.
- Solid knowledge of polymer science and crystal science.
- Familiar with electron beam welding techniques.
- Experienced in defect analysis using laser microscopy.

## INSTRUMENTAL SKILLS

### **Materials Characterization and Spectroscopy**

- Fourier Transform Infrared Spectroscopy (FTIR)
- X-ray Diffraction (XRD)
- X-ray Absorption Spectroscopy (XAS)
- X-ray Absorption Fine Structure (XAFS)
- Raman Spectroscopy
- UV–Visible Spectroscopy (UV–Vis)
- Photoluminescence Spectroscopy (PL)

### **Microscopy and Surface Analysis**

- Scanning Electron Microscopy (SEM)
- Transmission Electron Microscopy (TEM)
- Atomic Force Microscopy (AFM)
- Energy-Dispersive X-ray Spectroscopy (EDS)
- Laser Microscopy

### **Thermal and Physicochemical Analysis**

- Thermogravimetric Analysis (TGA)
- High-Performance Liquid Chromatography (HPLC)
- Gas Chromatography (GC)
- Density Measurement and Analysis
- pH Measurement and Chemical Analysis

### **Mechanical and Materials Testing**

- Hydraulic Tensile Testing
- Rockwell Hardness Testing
- Vickers Hardness Testing
- Pressure Leak Testing
- Coordinate Measuring Machine (CMM)

### **Materials Processing and Sample Preparation**

- Ball Milling
- Sample Cutting and Polishing
- Furnace and High-Temperature Heat Treatment
- Oven Drying and Thermal Processing
- Hot Plate with Magnetic Stirrer
- Water Bath Operations

### **General Laboratory Equipment and Techniques**

- Electronic Analytical Balance
- Dean-Stark Apparatus
- Dissolution Tester
- Melting Point Tester
- Disintegration Tester
- Standard Laboratory Safety and Experimental Procedures

## TECHNICAL SKILLS IN COMPUTATIONAL AND ANALYTICAL SOFTWARE

- |                                               |                                               |
|-----------------------------------------------|-----------------------------------------------|
| • Python                                      | • HighScore Plus                              |
| • Statistical Data Analysis and Visualization | • ImageJ                                      |
| • Machine Learning (ML)                       | • Management Software                         |
| • Artificial Intelligence (AI)                | • MS. Word, MS. Excel                         |
| • Materials Studio (CASTEP-DFT)               | • MS. PowerPoint, MS. Access                  |
| • OriginPro/OriginLab                         | • Auto CAD, SolidWorks                        |
| • VESTA application                           | • ANSYS                                       |
| • Mendeley                                    | • Windows, Linux, and macOS Operating Systems |
| • Plots 32                                    | • Programming C                               |
| • FullProf Suite                              | • Adobe Photoshop                             |

**MAJOR COURSES**

| <b>Doctoral course</b>                               | <b>Master`s course</b>                                           | <b>Bachelor course</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|------------------------------------------------------|------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Advanced Inorganic Materials Chemistry               | Advanced Engineering Materials,<br>Nano Technology,              | Introduction to Materials Science & Engineering,<br>Algebra, Trigonometry & Vector Analysis,<br>Mechanics & General Properties of Matter,<br>Waves and Sound,<br>Physical and Inorganic Chemistry,<br>Technical English,<br>Qualitative and Quantitative Analysis of Materials <b>(Lab)</b> ,<br>Physical Chemistry <b>(Lab)</b> ,<br>Mechanical Engineering Workshop <b>(Lab)</b> ,                                                                                                                                                          |
| Advanced physical properties of inorganic Materials, | Chemistry and Texture of Textiles,<br>Thin Films,                | Crystallography and Structure of Materials<br>Applied Electricity & Magnetism<br>Differential and Integral Calculus<br>Statistics for Engineers<br>Crystallography <b>(Lab)</b><br>Engineering Drawing and Graphics <b>(Lab)</b><br><br>Polymeric Materials<br>Basic Quantum Mechanics<br>Electronic Properties of Materials<br>Matrices and Differential Equations<br>Basic Electronics and Instrumentations<br>Industrial Management and Accountancy<br>Polymer Synthesis and Characterisation <b>(Lab)</b><br>General Physics <b>(Lab)</b> |
| Advance polymer materials chemistry                  | Advanced Ceramics,                                               | Materials Thermodynamics and Kinetics<br>Materials for Energy Conversion and Storage<br>Fundamentals of Electrical Engineering<br>Computer Operating Systems and Programming in C & C++<br>Organic Chemistry<br>Law and Professional Ethics<br>Computer Programming <b>(Lab)</b><br>General Chemistry <b>(Lab)</b>                                                                                                                                                                                                                            |
| Advance inorganic materials design                   | Advanced Magnetic Materials,                                     | Phase Diagram and Microstructure of Materials<br>Glass and Ceramics<br>Composite Materials<br>Mechanical Behaviour of Materials<br>Production of Metals<br>Glass and Ceramic Processing (Lab)<br>Mechanical Property Testing <b>(Lab)</b><br>Metallography and Microstructure <b>(Lab)</b>                                                                                                                                                                                                                                                    |
| Magnetic, Dielectric and Semiconducting Materials    | Biodegradable Polymers,<br>Extractive Metallurgy,                | Physical Metallurgy,<br>Materials Manufacturing Engineering<br>Corrosion Science and Engineering<br>Electrochemical Science and Engineering<br>Optics and Optical Properties of Materials<br>Electrochemical and Corrosion <b>(Lab)</b><br>Industrial Project <b>(Lab)</b>                                                                                                                                                                                                                                                                    |
| Energy and materials science course                  | Photonic Materials and Devices,<br>Advanced Nuclear Engineering, | Microscopic Methods in Materials Characterization<br>Surface Science and Engineering<br>Plastic and Rubber Technology<br>Magnetic, Dielectric and Semiconducting Materials<br>Construction Materials<br>Metallurgical <b>(Lab)</b>                                                                                                                                                                                                                                                                                                            |



|                                            |                                                |                                                                                                                                                                                                                                                                                                                                      |
|--------------------------------------------|------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                            |                                                | Polymer Processing ( <b>Lab</b> )<br>Electronic Materials ( <b>Lab</b> )                                                                                                                                                                                                                                                             |
| Crystallography,<br>Scientist Ethics, etc. | Functional Polymers,<br>Advanced Biomaterials, | Introduction to Nano and Biomaterials<br>X-Ray and Spectroscopic Methods in Materials Analysis<br>Welding and Joining Technology<br>Selection of Engineering Materials<br>Fibre Technology<br>Welding and Joining ( <b>Lab</b> )<br>Computer Aided Engineering Drawing & Designing ( <b>Lab</b> )<br>Research Project ( <b>Lab</b> ) |

#### LANGUAGE

- English (Fluent)
- Japanese (Moderate)
- Bengali (Native)
- Hindi (Fluent)
- Urdu (Fluent)
- Arabic (Basic)

#### INDUSTRIAL TRAINING/INTERNSHIP

|      |                                                                                |
|------|--------------------------------------------------------------------------------|
| 2025 | Mitsubishi Electric Corporation, Japan                                         |
| 2025 | Tada Electronic Co. Ltd., Japan                                                |
| 2025 | Tosei Electronbeam Co. Ltd., Japan                                             |
| 2023 | Sannichi YBS Group, Yamanashi, Japan                                           |
| 2023 | Nippo Precision Co.Ltd., Yamanashi, Japan                                      |
| 2021 | Panchagarh Sugar Mills Ltd.                                                    |
| 2014 | Sopura Silk Mills Ltd, Rajshahi, Bangladesh                                    |
| 2014 | Shinepukur Ceramics Limited (SCL), BEXIMCO Industrial Park, Dhaka, Bangladesh. |

#### EXTRA-CURRICULAR ACTIVITIES

- Active member of the Debate Team, GOLD Bangladesh, University of Rajshahi.
- Member of the Rajshahi University Higher Study Club (RUHSC).
- Member of the Rajshahi University Engineering Club (RUEC).
- Member of Badhon, a voluntary blood donation organization.
- Successfully completed a cultural workshop organized by Rajshahi University Shanskritik Jote.
- Served for two years as a Cultural Organizer with Bissobiddaloy Theater Rajshahi (BTR), a university-based cultural organization.
- Participated in student-led engineering and academic engagement activities.

#### Interests:

Reading research papers, mentoring and sharing life experiences with students, traveling, photography, shopping, cricket, and badminton.

#### REFERENCE

##### **Professor Dr. Takahiro Takei**

Center for Crystal science and Technology,  
University of Yamanashi, Japan  
E-mail: [takei@yamanashi.ac.jp](mailto:takei@yamanashi.ac.jp)

##### **Associate Professor Dr. Emrul Kayesh**

Department of Materials Science and Engineering  
University of Rajshahi, Bangladesh  
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##### **Professor Dr. Mirza Humaun Kabir Rubel**

Department of Materials Science and Engineering  
University of Rajshahi, Bangladesh  
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##### **Professor Dr. Md. Abdul Matin**

Department of Materials Science and Engineering  
University of Rajshahi, Bangladesh  
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##### **Professor Dr. Abu Mahmud**

Department of Materials Science and Engineering  
University of Rajshahi, Bangladesh  
Email: [mahmud.matsc@ru.ac.bd](mailto:mahmud.matsc@ru.ac.bd)

##### **Dr. Muhammad Abid Khan**

Specially Appointed Researcher  
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